Index laws

1 Write the following expressions in simplest index form:

a $2^{12} \times 2^{9}$

b $2^{8} \times 11^{8}$

c $11^{12} \div 11^{8}$

d $21^{5} \div 3^{5}$

e $\left(5^{12}\right)^{4}$

f $15^{17} \div 15^{8} \div 15^{5}$

g $\left(23^{8}\right)^{9} \times 23^{7}$

h $\dfrac{\left(17^{5}\right)^{8}}{17^{32}}$

i $\dfrac{19^{9} \times 19^{4}}{19^{8}}$

j $\dfrac{12^{6}}{12^{4}} \times 12^{5}$

k $\dfrac{\left(13^{5}\right)^{2} \times 13^{3}}{13^{5}}$

l $\dfrac{\left(15^{9}\right)^{5} \times 15^{7}}{15^{25}}$

2 Complete the following statements:

a $11^{11} \times 11^{\square} = 11^{19}$

b $5^{11} \times (\square)^{11} = 35^{11}$

c $97^{\square} \div 97^{22} = 97^{12}$

d $55^{6} \div (\square)^{6} = 11^{6}$

e $\left(11^{4}\right)^{\square} = 11^{12}$

f $7^{8} \times 7^{\square} = 7^{14}$

g $2^{10} \times (\square)^{10} = 10^{10}$

h $19^{\square} \div 19^{18} = 19^{20}$

i $60^{4} \div (\square)^{4} = 12^{4}$

j $\left(13^{8}\right)^{\square} = 13^{16}$

3 Evaluate the following expressions:

a $6^{5} \times 6^{3}$

b $7^{3} \times 3^{3}$

c $4^{8} \div 4^{3}$

d $\left(5^{4}\right)^{2}$

e $35^{5} \div 5^{5}$

f $2^{4} \times 4^{4}$

g $11^{18} \div 11^{9} \div 11^{7}$

h $\left(3^{3}\right)^{2}$

i $\dfrac{6^{5} \times 6^{9}}{6^{12}}$

j $7^{27} \div 7^{30} \div 7^{3}$

k $\dfrac{12^{10} \times 12^{4}}{12^{11}}$

l $\dfrac{\left(6^{8}\right)^{6}}{6^{46}}$

Negative bases

4 Write the following expressions in simplest index form:

a $(-11)^{10} \times (-11)^{3}$

b $(-7)^{8} \times 3^{8}$

c $(-5)^{2} \times 3^{2}$

d $(-3)^{12} \div (-3)^{5}$

e $(-12)^{20} \div (-12)^{19}$

f $(-30)^{50} \div (-30)^{47}$

g $(-48)^{3} \div (-6)^{3}$

h $(-33)^{11} \div (-3)^{11}$

i $(-35)^{5} \div 5^{5}$

j $(-42)^{2} \div 7^{2}$

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5 Complete the following statements:

a $11^3 \times (\square)^3 = (-77)^3$

b $(-5)^{11} \times (\square)^{11} = 15^{11}$

c $(-5)^{\square} \div (-5)^{31} = (-5)^{19}$

d $(-14)^{13} \div (\square)^{13} = (-7)^{13}$

e $(-3)^{\square} \div (-3)^{39} = (-3)^{10}$

f $(-5)^4 \times (\square)^4 = (-60)^4$

g $(-3)^7 \times (\square)^7 = (6)^7$

h $(\square)^3 \div (-3)^3 = (7)^3$

i $(-33)^3 \div (\square)^3 = -11^3$

j $(\square)^7 \div (-5)^7 = 11^7$

6 Evaluate the following expressions:

a $(-4)^{11} \div (-4)^7$

b $(-2)^3 \times (-2)^3$

c $(-3)^3 \times (-3)^2$

d $4^3 \times (-5)^3$

e $(-3)^8 \div (-3)^5$

f $15^5 \div (-3)^5$

g $2^3 \times (-3)^3$

h $(-14)^{11} \div 2^{11}$

i $(-7)^2 \times 5^2$

j $(-9)^4 \times (-3)^4$

k $(-100)^6 \div 50^6$

l $60^3 \div (-3)^3$

Fractional bases

7 Write the following in simplest index form:

a $\left(\frac{1}{3}\right)^4$

b $\left(\frac{3}{8}\right)^3$

c $\left(\frac{4}{16}\right)^8$

d $\left(\frac{15}{6}\right)^2$

e $\left(\frac{10}{33}\right)^5$

f $\left(\frac{2}{35}\right)^6$

g $\left(\frac{5}{18}\right)^3$

h $\left(\frac{29}{41}\right)^7$

i $\left(\frac{11}{13}\right)^9$

j $\left(\frac{20}{3}\right)^2$

k $\left(\frac{17}{4}\right)^4$

l $\left(\frac{31}{50}\right)^5$

8 Complete the following statements:

a $\frac{1}{27} = \left(\frac{1}{3}\right)^{\square}$

b $\frac{64}{27} = \left(\frac{4}{3}\right)^{\square}$

c $\frac{27}{8} = \left(\frac{3}{\square}\right)^3$

d $\frac{\square}{16} = \left(\frac{1}{4}\right)^2$

e $\frac{27}{8} = \left(\frac{3}{2}\right)^{\square}$

f $\frac{\square}{625} = \left(\frac{2}{5}\right)^4$

g $\frac{81}{100} = \left(\frac{\square}{10}\right)^2$

h $\frac{256}{\square} = \left(\frac{4}{5}\right)^4$

i $\frac{25}{9} = \left(\frac{5}{3}\right)^{\square}$

j $\frac{\square}{144} = \left(\frac{11}{12}\right)^2$

k $\frac{125}{27} = \left(\frac{\square}{3}\right)^3$

l $\frac{32}{729} = \left(\frac{2}{3}\right)^{\square}$

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9 Evaluate the following expressions in fully simplified fraction:

a $\left(\frac{1}{3}\right)^2$ **b** $\left(\frac{3}{5}\right)^3$ **c** $\left(\frac{1}{2}\right)^5$ **d** $\left(\frac{7}{11}\right)^2$

e $\left(\frac{3}{8}\right)^3$ **f** $\left(\frac{12}{13}\right)^2$ **g** $\left(\frac{5}{9}\right)^3$ **h** $\left(\frac{2}{3}\right)^4$

i $\left(\frac{4}{5}\right)^6$ **j** $\left(\frac{2}{5}\right)^3$ **k** $\left(\frac{5}{7}\right)^2$ **l** $\left(\frac{10}{14}\right)^3$

10 State whether the following fractions are equal to $\left(\frac{1}{2}\right)^3$:

a $\frac{3}{6}$ **b** $\frac{1}{2^3}$ **c** $\frac{3}{2^3}$ **d** $\frac{1}{8}$

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